

PANDUAN LENGKAP PEMBANGUNAN

Sistem Informasi Manajemen & Monitoring Produksi & Penjualan Telur Ayam Petelur

Hans Jaya Poultry Farm

Tech Stack Laravel 12 Breeze + Blade Tailwind CSS v4 Alpine.js Chart.js MySQL 8	Package Tambahan Spatie Permission Barryvdh DomPDF Maatwebsite Excel Laravel Vite npm / Node.js Laragon / Herd
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BAB 1 — Persiapan Lingkungan & Instalasi

Sebelum mulai coding, pastikan semua tools berikut sudah terinstal di komputer.

1.1 Kebutuhan Sistem

PHP	Versi 8.2 atau lebih baru (wajib untuk Laravel 12)
Composer	Versi 2.x — package manager PHP
Node.js	Versi 18 atau lebih baru (untuk Vite & Tailwind)
npm	Sudah bundled bersama Node.js
MySQL	Versi 8.x
Web Server	Laragon (Windows) atau Laravel Herd (Mac) atau XAMPP
Editor	VS Code — recommended dengan extension PHP Intelephense

💡 Cek versi PHP: `php -v` | Cek Composer: `composer -V` | Cek Node: `node -v`

1.2 Cek Versi di Terminal

```
# Cek semua versi sekaligus
php -v
composer -V
node -v
npm -v
mysql --version
```

1.3 Install Laravel 12

Buka terminal, arahkan ke folder htdocs (Laragon) atau www, lalu jalankan:

```
# Buat project Laravel 12 baru
composer create-project laravel/laravel hans-jaya-poultry

# Masuk ke folder project
cd hans-jaya-poultry

# Cek versi Laravel (pastikan 12.x)
php artisan --version
```

1.4 Install Laravel Breeze

Breeze menyediakan autentikasi (login, register, logout) + setup Tailwind CSS v4 secara otomatis.

```
# Install Breeze via Composer
composer require laravel/breeze --dev

# Install dengan stack Blade (bukan React/Vue)
php artisan breeze:install blade

# Install dependencies Node.js
npm install

# Build assets (Tailwind, Alpine.js)
npm run dev
```

 Setelah perintah `breeze:install blade`, Tailwind CSS v4 dan Alpine.js sudah otomatis terkonfigurasi.

1.5 Konfigurasi Database

Buka file `.env` di root project, sesuaikan konfigurasi database:

```
APP_NAME="Hans Jaya Poultry Farm"
APP_ENV=local
APP_DEBUG=true
APP_URL=http://localhost

DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_PORT=3306
DB_DATABASE=hans_jaya_poultry
DB_USERNAME=root
DB_PASSWORD=
```

Buat database di MySQL:

```
CREATE DATABASE hans_jaya_poultry CHARACTER SET utf8mb4 COLLATE
utf8mb4_unicode_ci;
```

1.6 Install Package Tambahan

```
# Role & Permission management
composer require spatie/laravel-permission

# Export PDF
composer require barryvdh/laravel-dompdf

# Export Excel
composer require maatwebsite/excel

# Publish config Spatie
php artisan vendor:publish --
provider="Spatie\Permission\PermissionServiceProvider"
```

BAB 2 — Migration Database (8 Tabel)

Buat semua file migration secara berurutan. Urutan penting karena ada relasi foreign key.

2.1 Buat File Migration

```
php artisan make:migration create_kandang_table
php artisan make:migration add_kandang_id_to_users_table
php artisan make:migration create_produksi_telur_table
php artisan make:migration create_harga_telur_table
php artisan make:migration create_penjualan_table
php artisan make:migration create_detail_penjualan_table
php artisan make:migration create_stok_telur_table
php artisan make:migration create_pengaturan_table
```

2.2 Migration: kandang

```
Schema::create('kandang', function (Blueprint $table) {
    $table->id();
    $table->string('nama_kandang', 100);
    $table->unsignedInteger('jumlah_ayam')->default(0);
    $table->text('keterangan')->nullable();
    $table->enum('status', ['aktif', 'nonaktif'])->default('aktif');
    $table->timestamps();
});
```

2.3 Migration: tambah kandang_id ke users

```
Schema::table('users', function (Blueprint $table) {
```

```

$table->enum('role', ['pemilik', 'pekerja'])->after('password');
$table->foreignId('kandang_id')
    ->nullable()
    ->after('role')
    ->constrained('kandang')
    ->nullOnDelete();
});

```

2.4 Migration: produksi_telur

```

Schema::create('produksi_telur', function (Blueprint $table) {
    $table->id();
    $table->foreignId('kandang_id')->constrained('kandang')->
    >cascadeOnDelete();
    $table->foreignId('user_id')->constrained('users')->cascadeOnDelete();
    $table->date('tanggal_produksi');
    $table->enum('satuan_input', ['butir', 'kg']);
    $table->decimal('jumlah_input', 10, 2);
    $table->unsignedInteger('jumlah_butir')->default(0);
    $table->decimal('jumlah_kg', 10, 3)->default(0);
    $table->text('keterangan')->nullable();
    $table->timestamps();
});

```

2.5 Migration: harga_telur

```

Schema::create('harga_telur', function (Blueprint $table) {
    $table->id();
    $table->enum('jenis_harga', ['kandang', 'grosir', 'konsumen']);
    $table->decimal('harga_per_kg', 12, 2);
    $table->decimal('harga_per_butir', 12, 2)->nullable();
    $table->date('tanggal_berlaku');
    $table->foreignId('user_id')->constrained('users')->cascadeOnDelete();
    $table->text('keterangan')->nullable();
    $table->timestamp('created_at')->useCurrent();
});

```

2.6 Migration: penjualan

```

Schema::create('penjualan', function (Blueprint $table) {
    $table->id();
    $table->foreignId('user_id')->constrained('users')->cascadeOnDelete();
    $table->date('tanggal_jual');
    $table->string('nama_pembeli', 100)->nullable();
    $table->decimal('total_harga', 15, 2)->default(0);
    $table->text('keterangan')->nullable();
    $table->timestamps();
});

```

2.7 Migration: detail_penjualan

```

Schema::create('detail_penjualan', function (Blueprint $table) {
    $table->id();

```

```

    $table->foreignId('penjualan_id')->constrained('penjualan')->cascadeOnDelete();
    $table->foreignId('harga_telur_id')->constrained('harga_telur');
    $table->enum('satuan_jual', ['butir', 'kg']);
    $table->decimal('jumlah_jual', 10, 2);
    $table->unsignedInteger('jumlah_butir')->default(0);
    $table->decimal('jumlah_kg', 10, 3)->default(0);
    $table->decimal('harga_satuan', 12, 2);
    $table->decimal('subtotal', 15, 2)->default(0);
    $table->timestamp('created_at')->useCurrent();
});

```

2.8 Migration: stok_telur

```

Schema::create('stok_telur', function (Blueprint $table) {
    $table->id();
    $table->unsignedInteger('stok_butir')->default(0);
    $table->decimal('stok_kg', 10, 3)->default(0);
    $table->timestamp('updated_at')->useCurrent();
});

```

2.9 Migration: pengaturan

```

Schema::create('pengaturan', function (Blueprint $table) {
    $table->id();
    $table->string('kunci', 100)->unique();
    $table->string('nilai', 255);
    $table->enum('tipe_data', ['string', 'integer', 'decimal', 'boolean'])
        ->default('string');
    $table->text('keterangan')->nullable();
    $table->timestamp('updated_at')->useCurrent();
});

```

2.10 Jalankan Migration

```
php artisan migrate
```

 Jalankan setelah semua file migration dibuat. Urutan otomatis berdasarkan timestamp file.

BAB 3 — Model Eloquent & Relasi

Setiap tabel memiliki Model. Model mendefinisikan relasi antar tabel menggunakan Eloquent ORM.

3.1 Buat Semua Model

```

php artisan make:model Kandang
php artisan make:model ProduksiTelur
php artisan make:model HargaTelur
php artisan make:model Penjualan

```

```
php artisan make:model DetailPenjualan
php artisan make:model StokTelur
php artisan make:model Pengaturan
```

3.2 Model User (update existing)

```
// app/Models/User.php
use Spatie\Permission\Traits\HasRoles;

class User extends Authenticatable {
    use HasRoles;

    protected $fillable = ['name', 'username', 'password', 'role',
        'kandang_id'];

    public function kandang() {
        return $this->belongsTo(Kandang::class);
    }
    public function produksiTelur() {
        return $this->hasMany(ProduksiTelur::class);
    }
    public function penjualan() {
        return $this->hasMany(Penjualan::class);
    }
}
```

3.3 Model Kandang

```
// app/Models/Kandang.php
class Kandang extends Model {
    protected $table = 'kandang';
    protected $fillable = ['nama_kandang', 'jumlah_ayam', 'keterangan',
        'status'];

    public function produksiTelur() {
        return $this->hasMany(ProduksiTelur::class);
    }
    public function pekerja() {
        return $this->hasMany(User::class);
    }
}
```

3.4 Model ProduksiTelur

```
// app/Models/ProduksiTelur.php
class ProduksiTelur extends Model {
    protected $table = 'produksi_telur';
    protected $fillable = [
        'kandang_id', 'user_id', 'tanggal_produksi',
        'satuan_input', 'jumlah_input', 'jumlah_butir', 'jumlah_kg',
        'keterangan'
    ];

    public function kandang() { return $this->belongsTo(Kandang::class); }
```

```
public function user()    { return $this->belongsTo(User::class); }
}
```

3.5 Model Penjualan & DetailPenjualan

```
// app/Models/Penjualan.php
class Penjualan extends Model {
    protected $fillable =
    ['user_id', 'tanggal_jual', 'nama_pembeli', 'total_harga', 'keterangan'];

    public function user()    { return $this->belongsTo(User::class); }
    public function detail() { return $this->hasMany(DetailPenjualan::class); }
}

// app/Models/DetailPenjualan.php
class DetailPenjualan extends Model {
    protected $table = 'detail_penjualan';
    protected $fillable = [
        'penjualan_id', 'harga_telur_id', 'satuan_jual',
        'jumlah_jual', 'jumlah_butir', 'jumlah_kg', 'harga_satuan', 'subtotal'
    ];

    public function penjualan() { return $this->belongsTo(Penjualan::class); }
}
    public function hargaTelur() { return $this->belongsTo(HargaTelur::class); }
}
}
```

BAB 4 — Role & Permission (Spatie)

Sistem ini memiliki 2 role: pemilik (akses penuh) dan pekerja (hanya input produksi kandangnya sendiri).

4.1 Setup Spatie Permission

```
# Publish migration Spatie
php artisan vendor:publish --
provider="Spatie\Permission\PermissionServiceProvider"

# Jalankan migration Spatie
php artisan migrate
```

4.2 Tambahkan HasRoles ke Model User

```
use Spatie\Permission\Traits\HasRoles;

class User extends Authenticatable {
    use HasFactory, Notifiable, HasRoles;
    // ...
}
```

```
}
```

4.3 Buat Seeder Role & User

```
php artisan make:seeder RoleSeeder
php artisan make:seeder UserSeeder

// database/seeder/RoleSeeder.php
use Spatie\Permission\Models\Role;

public function run() {
    Role::create(['name' => 'pemilik', 'guard_name' => 'web']);
    Role::create(['name' => 'pekerja', 'guard_name' => 'web']);
}

// database/seeder/UserSeeder.php
public function run() {
    $pemilik = User::create([
        'name' => 'Hans Jaya',
        'email' => 'pemilik@hansjaya.com',
        'password' => bcrypt('password123'),
        'role' => 'pemilik',
    ]);
    $pemilik->assignRole('pemilik');
}

php artisan db:seed --class=RoleSeeder
php artisan db:seed --class=UserSeeder
```

4.4 Middleware Role di Route

```
// routes/web.php

// Route khusus pemilik
Route::middleware(['auth', 'role:pemilik'])->group(function () {
    Route::resource('kandang', KandangController::class);
    Route::resource('harga', HargaTelurController::class);
    Route::resource('penjualan', PenjualanController::class);
    Route::get('laporan/produksi', [LaporanController::class, 'produksi']);
    Route::get('laporan/penjualan', [LaporanController::class, 'penjualan']);
    Route::resource('pengaturan', PengaturanController::class);
});

// Route khusus pekerja
Route::middleware(['auth', 'role:pekerja'])->group(function () {
    Route::resource('produksi', ProduksiTelurController::class)-
>only(['index', 'create', 'store']);
});

// Dashboard (semua role)
Route::middleware(['auth'])->group(function () {
    Route::get('/dashboard', [DashboardController::class, 'index'])->
name('dashboard');
});
```


BAB 5 — Controller & Logika CRUD

5.1 Buat Semua Controller

```
php artisan make:controller DashboardController
php artisan make:controller KandangController --resource
php artisan make:controller ProduksiTelurController --resource
php artisan make:controller HargaTelurController --resource
php artisan make:controller PenjualanController --resource
php artisan make:controller LaporanController
php artisan make:controller PengaturanController --resource
```

5.2 Logika Konversi Otomatis (ProduksiTelurController)

Konversi butir ↔ kg mengambil nilai dari tabel pengaturan, bukan hardcode.

```
// app/Http/Controllers/ProduksiTelurController.php

public function store(Request $request) {
    $request->validate([
        'tanggal_produksi' => 'required|date',
        'satuan_input'      => 'required|in:butir,kg',
        'jumlah_input'     => 'required|numeric|min:0',
    ]);

    // Ambil nilai konversi dari tabel pengaturan
    $konversi = Pengaturan::where('kunci', 'konversi_butir_per_kg')->value('nilai');
    $konversi = (float) $konversi ?: 16;

    if ($request->satuan_input === 'butir') {
        $jumlah_butir = (int) $request->jumlah_input;
        $jumlah_kg    = round($jumlah_butir / $konversi, 3);
    } else {
        $jumlah_kg    = (float) $request->jumlah_input;
        $jumlah_butir = (int) round($jumlah_kg * $konversi);
    }

    ProduksiTelur::create([
        'kandang_id' => auth()->user()->kandang_id,
        'user_id'    => auth()->id(),
        'tanggal_produksi' => $request->tanggal_produksi,
        'satuan_input'    => $request->satuan_input,
        'jumlah_input'    => $request->jumlah_input,
        'jumlah_butir'    => $jumlah_butir,
        'jumlah_kg'       => $jumlah_kg,
        'keterangan'     => $request->keterangan,
    ]);

    // Update stok otomatis
    $stok = StokTelur::first();
    $stok->increment('stok_butir', $jumlah_butir);
}
```

```

$stok->stok_kg = round($stok->stok_butir / $konversi, 3);
$stok->save();

return redirect()->route('produksi.index')
    ->with('success', 'Data produksi berhasil disimpan!');
}

```

5.3 Logika Penjualan (Update Stok Otomatis)

```

// app/Http/Controllers/PenjualanController.php

public function store(Request $request) {
    DB::transaction(function () use ($request) {
        $penjualan = Penjualan::create([
            'user_id'      => auth()->id(),
            'tanggal_jual' => $request->tanggal_jual,
            'nama_pembeli' => $request->nama_pembeli,
            'total_harga'  => 0,
        ]);

        $total = 0;
        $totalButir = 0;
        $konversi = (float) Pengaturan::where('kunci',
            'konversi_butir_per_kg')->value('nilai') ?: 16;

        foreach ($request->items as $item) {
            $harga = HargaTelur::findOrFail($item['harga_telur_id']);
            $hargaSatuan = $item['satuan_jual'] === 'kg'
                ? $harga->harga_per_kg
                : $harga->harga_per_butir;

            $jumlahButir = $item['satuan_jual'] === 'butir'
                ? (int) $item['jumlah_jual']
                : (int) round($item['jumlah_jual'] * $konversi);

            $subtotal = $item['jumlah_jual'] * $hargaSatuan;
            $total += $subtotal;
            $totalButir += $jumlahButir;

            DetailPenjualan::create([
                'penjualan_id' => $penjualan->id,
                'harga_telur_id' => $item['harga_telur_id'],
                'satuan_jual' => $item['satuan_jual'],
                'jumlah_jual' => $item['jumlah_jual'],
                'jumlah_butir' => $jumlahButir,
                'jumlah_kg' => round($jumlahButir / $konversi, 3),
                'harga_satuan' => $hargaSatuan,
                'subtotal' => $subtotal,
            ]);
        }

        $penjualan->update(['total_harga' => $total]);

        // Kurangi stok
        $stok = StokTelur::first();
        $stok->decrement('stok_butir', $totalButir);
        $stok->stok_kg = round($stok->stok_butir / $konversi, 3);
    });
}

```

```

        $stok->save();
    });

    return redirect()->route('penjualan.index')
        ->with('success', 'Transaksi penjualan berhasil
disimpan!');
}

```

BAB 6 — Layout Blade & Tailwind CSS v4

6.1 Struktur Folder Views

```

resources/views/
├── auth/                → login, register (dari Breeze)
├── layouts/
│   ├── app.blade.php    → layout utama (sidebar + navbar)
│   └── guest.blade.php  → layout halaman auth
├── components/
│   ├── sidebar.blade.php
│   ├── navbar.blade.php
│   ├── card.blade.php
│   └── alert.blade.php
├── dashboard/index.blade.php
├── kandang/
├── produksi/
├── harga/
├── penjualan/
├── stok/
├── laporan/
└── pengaturan/

```

6.2 Layout Utama (app.blade.php)

```

<!DOCTYPE html>
<html lang="id">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>{{ config('app.name') }}</title>
    @vite(['resources/css/app.css', 'resources/js/app.js'])
</head>
<body class="bg-gray-100 font-sans">
    <div class="flex h-screen overflow-hidden">

        <!-- Sidebar -->
        <x-sidebar />

        <!-- Main Content -->
        <div class="flex-1 flex flex-col overflow-auto">
            <x-navbar />
            <main class="flex-1 p-6">
                <!-- Flash Message -->
            </main>
        </div>
    </div>

```

```

        @if(session('success'))
            <div class="mb-4 p-4 bg-green-100 text-green-800 rounded-
lg">
                {{ session('success') }}
            </div>
        @endif
        {{ $slot }}
    </main>
</div>
</div>
</body>
</html>

```

6.3 Sidebar Component

```

<!-- resources/views/components/sidebar.blade.php -->
<div class="w-64 bg-blue-900 text-white flex flex-col min-h-screen">
    <!-- Logo -->
    <div class="p-6 border-b border-blue-700">
        <h1 class="text-lg font-bold">Hans Jaya</h1>
        <p class="text-blue-300 text-sm">Poultry Farm</p>
    </div>

    <!-- Menu -->
    <nav class="flex-1 p-4 space-y-1">
        <a href="{{ route('dashboard') }}"
            class="flex items-center gap-3 px-4 py-3 rounded-lg hover:bg-blue-
700 transition">
            Dashboard
        </a>
        @role('pemilik')
        <a href="{{ route('kandang.index') }}"
            class="flex items-center gap-3 px-4 py-3 rounded-lg hover:bg-blue-
700 transition">
            Kandang
        </a>
        @endrole
        <a href="{{ route('produksi.index') }}"
            class="flex items-center gap-3 px-4 py-3 rounded-lg hover:bg-blue-
700 transition">
            Produksi Telur
        </a>
        @role('pemilik')
        <a href="{{ route('penjualan.index') }}"
            class="flex items-center gap-3 px-4 py-3 rounded-lg hover:bg-blue-
700 transition">
            Penjualan
        </a>
        <a href="{{ route('laporan.produksi') }}"
            class="flex items-center gap-3 px-4 py-3 rounded-lg hover:bg-blue-
700 transition">
            Laporan
        </a>
        @endrole
    </nav>
</div>

```

6.4 Contoh Halaman CRUD (Tabel dengan Tailwind)

```

<!-- resources/views/kandang/index.blade.php -->
<x-app-layout>
    <div class="bg-white rounded-xl shadow p-6">
        <div class="flex justify-between items-center mb-6">
            <h2 class="text-xl font-bold text-gray-800">Data Kandang</h2>
            <a href="{{ route('kandang.create') }}"
                class="bg-blue-600 text-white px-4 py-2 rounded-lg hover:bg-
blue-700">
                + Tambah Kandang
            </a>
        </div>

        <table class="w-full text-sm text-left">
            <thead class="bg-blue-50 text-blue-800 uppercase text-xs">
                <tr>
                    <th class="px-4 py-3">No</th>
                    <th class="px-4 py-3">Nama Kandang</th>
                    <th class="px-4 py-3">Jumlah Ayam</th>
                    <th class="px-4 py-3">Status</th>
                    <th class="px-4 py-3">Aksi</th>
                </tr>
            </thead>
            <tbody>
                @foreach($kandang as $k)
                    <tr class="border-b hover:bg-gray-50">
                        <td class="px-4 py-3">{{ $loop->iteration }}</td>
                        <td class="px-4 py-3 font-medium">{{ $k->nama_kandang
}}</td>
                        <td class="px-4 py-3">{{ number_format($k->jumlah_ayam) }}
ekor</td>
                        <td class="px-4 py-3">
                            <span class="px-2 py-1 rounded-full text-xs
                                {{ $k->status === 'aktif' ? 'bg-green-100 text-
green-700' : 'bg-red-100 text-red-700' }}">
                                    {{ ucfirst($k->status) }}
                                </span>
                        </td>
                        <td class="px-4 py-3 flex gap-2">
                            <a href="{{ route('kandang.edit', $k) }}"
                                class="text-blue-600 hover:underline">Edit</a>
                            <form method="POST" action="{{
route('kandang.destroy', $k) }}">
                                @csrf @method('DELETE')
                                <button class="text-red-500
hover:underline">Hapus</button>
                            </form>
                        </td>
                    </tr>
                @endforeach
            </tbody>
        </table>
    </div>
</x-app-layout>

```

BAB 7 — Grafik dengan Chart.js

7.1 Install Chart.js

```
# Via npm (recommended)
npm install chart.js

# Atau via CDN langsung di blade (alternatif simpel)
# Tambahkan di layout sebelum </body>:
# <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
```

7.2 Data Grafik di Controller

```
// app/Http/Controllers/LaporanController.php

public function produksi(Request $request) {
    $bulan = $request->bulan ?? now()->month;
    $tahun = $request->tahun ?? now()->year;

    // Data produksi per hari dalam bulan
    $data = ProduksiTelur::whereMonth('tanggal_produksi', $bulan)
        ->whereYear('tanggal_produksi', $tahun)
        ->selectRaw('DATE(tanggal_produksi) as tanggal, SUM(jumlah_butir) as
total')
        ->groupBy('tanggal')
        ->orderBy('tanggal')
        ->get();

    // Data per kandang
    $perKandang = ProduksiTelur::with('kandang')
        ->whereMonth('tanggal_produksi', $bulan)
        ->whereYear('tanggal_produksi', $tahun)
        ->selectRaw('kandang_id, SUM(jumlah_butir) as total')
        ->groupBy('kandang_id')
        ->get();

    return view('laporan.produksi', compact('data', 'perKandang', 'bulan',
'tahun'));
}
```

7.3 Grafik Garis — Produksi Harian

```
<!-- resources/views/laporan/produksi.blade.php -->
<x-app-layout>
    <div class="bg-white rounded-xl shadow p-6 mb-6">
        <h2 class="text-xl font-bold mb-4">Grafik Produksi Harian</h2>
        <canvas id="chartProduksi" height="100"></canvas>
    </div>

    <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
    <script>
        const labels = @json($data->pluck('tanggal'));
        const values = @json($data->pluck('total'));

        new Chart(document.getElementById('chartProduksi'), {
            type: 'line',
            data: {
                labels: labels,
```

```

        datasets: [{
          label: 'Jumlah Produksi (Butir)',
          data: values,
          borderColor: '#2E75B6',
          backgroundColor: 'rgba(46, 117, 182, 0.1)',
          fill: true,
          tension: 0.4,
          pointRadius: 4,
        }]
      },
      options: {
        responsive: true,
        plugins: {
          legend: { position: 'top' },
          title: { display: true, text: 'Produksi Telur Harian' }
        },
        scales: {
          y: { beginAtZero: true }
        }
      }
    });
  </script>
</x-app-layout>

```

7.4 Grafik Donut — Produksi per Kandang

```

<div class="bg-white rounded-xl shadow p-6">
  <h2 class="text-xl font-bold mb-4">Produksi per Kandang</h2>
  <canvas id="chartKandang" height="120"></canvas>
</div>

<script>
  const kandangLabels = @json($perKandang->map(fn($k) => $k->kandang->nama_kandang));
  const kandangData = @json($perKandang->pluck('total'));

  new Chart(document.getElementById('chartKandang'), {
    type: 'doughnut',
    data: {
      labels: kandangLabels,
      datasets: [{
        data: kandangData,
        backgroundColor: ['#2E75B6', '#1F4E79', '#70AD47', '#ED7D31'],
      }]
    },
    options: {
      responsive: true,
      plugins: {
        legend: { position: 'bottom' }
      }
    }
  });
</script>

```

BAB 8 — Export Laporan PDF & Excel

8.1 Export PDF dengan DomPDF

```
// app/Http/Controllers/LaporanController.php

use Barryvdh\DomPDF\Facade\Pdf;

public function exportPdf(Request $request) {
    $data = ProduksiTelur::with('kandang', 'user')
        ->whereMonth('tanggal_produksi', $request->bulan)
        ->whereYear('tanggal_produksi', $request->tahun)
        ->orderBy('tanggal_produksi')
        ->get();

    $pdf = Pdf::loadView('laporan.pdf.produksi', compact('data'))
        ->setPaper('a4', 'landscape');

    return $pdf->download('laporan-produksi.pdf');
}
```

8.2 View PDF (laporan/pdf/produksi.blade.php)

```
<!DOCTYPE html><html><head>
    <style>
        body { font-family: Arial, sans-serif; font-size: 12px; }
        h2 { color: #1F4E79; }
        table { width: 100%; border-collapse: collapse; }
        th { background: #2E75B6; color: white; padding: 8px; }
        td { border: 1px solid #ccc; padding: 6px; }
        tr:nth-child(even) { background: #f0f7ff; }
    </style>
</head><body>
    <h2>Laporan Produksi Telur — Hans Jaya Poultry Farm</h2>
    <table>
        <tr>
            <th>No</th><th>Tanggal</th><th>Kandang</th>
            <th>Satuan</th><th>Jumlah (Butir)</th><th>Jumlah (Kg)</th>
        </tr>
        @foreach($data as $i => $d)
            <tr>
                <td>{{ $i+1 }}</td>
                <td>{{ $d->tanggal_produksi }}</td>
                <td>{{ $d->kandang->nama_kandang }}</td>
                <td>{{ $d->satuan_input }}</td>
                <td>{{ number_format($d->jumlah_butir) }}</td>
                <td>{{ number_format($d->jumlah_kg, 3) }}</td>
            </tr>
        @endforeach
    </table>
</body></html>
```

8.3 Export Excel dengan Maatwebsite


```

php artisan make:export ProduksiExport --model=ProduksiTelur

// app/Exports/ProduksiExport.php
use Maatwebsite\Excel\Concerns\FromCollection;
use Maatwebsite\Excel\Concerns\WithHeadings;

class ProduksiExport implements FromCollection, WithHeadings {

    public function __construct(private $bulan, private $tahun) {}

    public function collection() {
        return ProduksiTelur::with('kandang')
            ->whereMonth('tanggal_produksi', $this->bulan)
            ->whereYear('tanggal_produksi', $this->tahun)
            ->get()->map(fn($d) => [
                'Tanggal' => $d->tanggal_produksi,
                'Kandang' => $d->kandang->nama_kandang,
                'Satuan' => $d->satuan_input,
                'Butir' => $d->jumlah_butir,
                'Kg' => $d->jumlah_kg,
            ]);
    }

    public function headings(): array {
        return ['Tanggal', 'Kandang', 'Satuan', 'Jumlah (Butir)', 'Jumlah (Kg)'];
    }
}

// Di Controller
use Maatwebsite\Excel\Facades\Excel;

public function exportExcel(Request $request) {
    return Excel::download(
        new ProduksiExport($request->bulan, $request->tahun),
        'laporan-produksi.xlsx'
    );
}

```

BAB 9 — Dashboard & Ringkasan Data

9.1 DashboardController

```

// app/Http/Controllers/DashboardController.php

public function index() {
    $stok = StokTelur::first();

    // Produksi hari ini
    $produksiHariIni = ProduksiTelur::whereDate('tanggal_produksi', today())
        ->sum('jumlah_butir');
}

```

```

// Penjualan bulan ini
$penjualanBulanIni = Penjualan::whereMonth('tanggal_jual', now()->month)
->whereYear('tanggal_jual', now()->year)
->sum('total_harga');

// Jumlah kandang aktif
$jumlahKandang = Kandang::where('status', 'aktif')->count();

// Grafik 7 hari terakhir
$produksi7Hari = ProduksiTelur::whereBetween('tanggal_produksi', [
    now()->subDays(6), now()
])
->selectRaw('DATE(tanggal_produksi) as tgl, SUM(jumlah_butir) as
total')
->groupBy('tgl')->orderBy('tgl')->get();

return view('dashboard.index', compact(
    'stok', 'produksiHariIni', 'penjualanBulanIni', 'jumlahKandang',
    'produksi7Hari'
));
}

```

9.2 Tampilan Dashboard (Card + Grafik)

```

<!-- resources/views/dashboard/index.blade.php -->
<x-app-layout>
    <!-- Summary Cards -->
    <div class="grid grid-cols-1 md:grid-cols-4 gap-4 mb-6">
        <div class="bg-white rounded-xl shadow p-5 border-l-4 border-blue-
500">
            <p class="text-gray-500 text-sm">Stok Telur</p>
            <p class="text-2xl font-bold text-blue-700">{{
number_format($stok->stok_butir) }}</p>
            <p class="text-sm text-gray-400">{{ number_format($stok->stok_kg,
1) }} kg</p>
        </div>
        <div class="bg-white rounded-xl shadow p-5 border-l-4 border-green-
500">
            <p class="text-gray-500 text-sm">Produksi Hari Ini</p>
            <p class="text-2xl font-bold text-green-700">{{
number_format($produksiHariIni) }}</p>
            <p class="text-sm text-gray-400">butir</p>
        </div>
        <div class="bg-white rounded-xl shadow p-5 border-l-4 border-yellow-
500">
            <p class="text-gray-500 text-sm">Penjualan Bulan Ini</p>
            <p class="text-2xl font-bold text-yellow-700">
                Rp {{ number_format($penjualanBulanIni) }}
            </p>
        </div>
        <div class="bg-white rounded-xl shadow p-5 border-l-4 border-purple-
500">
            <p class="text-gray-500 text-sm">Kandang Aktif</p>
            <p class="text-2xl font-bold text-purple-700">{{ $jumlahKandang
}}</p>
            <p class="text-sm text-gray-400">kandang</p>
        </div>
    </div>

```

```

<!-- Grafik 7 hari -->
<div class="bg-white rounded-xl shadow p-6">
  <h3 class="font-bold mb-4">Produksi 7 Hari Terakhir</h3>
  <canvas id="chartDashboard" height="80"></canvas>
</div>

<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
<script>
  new Chart(document.getElementById('chartDashboard'), {
    type: 'bar',
    data: {
      labels: @json($produksi7Hari->pluck('tgl')),
      datasets: [{
        label: 'Produksi (Butir)',
        data: @json($produksi7Hari->pluck('total')),
        backgroundColor: 'rgba(46, 117, 182, 0.7)',
        borderRadius: 6,
      }]
    },
    options: { responsive: true, scales: { y: { beginAtZero: true } } }
  }
  );
</script>
</x-app-layout>

```

BAB 10 — Urutan Pengerjaan Lengkap

Ikuti urutan ini agar proses pembangunan sistem berjalan sistematis dan tidak ada yang terlewat.

1

Setup Project

composer create-project + breeze:install + npm install

2

Konfigurasi .env

DB_DATABASE, DB_USERNAME, DB_PASSWORD

3

Install Package

Spatie Permission, DomPDF, Maatwebsite Excel

4

Buat Migration (8 tabel)

Urutan: kandang → users → produksi → harga → penjualan → detail → stok → pengaturan

5

Jalankan Migration

php artisan migrate

6

Buat Model + Relasi

Semua 8 model dengan method relasi Eloquent

7

Setup Role & Seeder

Buat role pemilik & pekerja, buat akun awal

8

Buat Controller

Semua controller dengan logika CRUD + konversi

9

Definisikan Route

Group middleware role:pemilik dan role:pekerja

10

Buat Layout & Views

app.blade.php, sidebar, navbar, semua halaman CRUD

11

Integrasi Chart.js

Grafik di halaman laporan dan dashboard

12

Export PDF & Excel

DomPDF untuk PDF, Maatwebsite untuk Excel

13

Testing & Bug Fix

Uji semua fitur dari kedua role, validasi form, cek stok

14

Polish & Dokumentasi

Rapikan tampilan, tambah notifikasi, tulis laporan TA